



**University of
Sunderland**

Faculty of Technology
Department of Computer Science

PROM02 – MSc Dissertation
MSc Data Science

Student Name: Freedom Ehikhamwen

Supervisor Name: Dr. Hazem Eissa

Second Marker Name: [Dr. Qiang Huang]

**Analysing Customer sentiment in social
media for product review (Movie Product)**

To create a sentiment analysis model that uses natural language processing (NLP) techniques to categorize product reviews and social media posts as positive, negative, or neutral; to extract important customer sentiments and themes; and to assess how sentiment insights can help improve customer satisfaction, direct product enhancements, and support business marketing campaigns.

[12/09/2025]

Declaration

I declare the following:

(1) that the material contained in this dissertation is the end result of my own work and that due acknowledgement has been given in the bibliography and references to **ALL** sources be they printed, electronic or personal.

(2) the Word Count of this Dissertation is [13,241]

(3) that unless this dissertation has been confirmed as confidential, I agree to an entire electronic copy or sections of the dissertation to being placed on the eLearning Portal, if deemed appropriate, to allow future students the opportunity to see examples of past dissertations. I understand that if displayed on the eLearning Portal it would be made available for no longer than five years and that students would be able to print off copies or download.

(4) I agree to my dissertation being submitted to a plagiarism detection service, where it will be stored in a database and compared against work submitted from this or any other Department or from other institutions using the service.

In the event of the service detecting a high degree of similarity between content within the service this will be reported back to my supervisor and second marker, who may decide to undertake further investigation that may ultimately lead to disciplinary actions, should instances of plagiarism be detected.

(5) I have read the University of Sunderland Policy Statement on Ethics in Research and I confirm that ethical issues have been considered, evaluated and appropriately addressed in this research.

SIGNED: Freedom Ehikhamwen

DATE: 12/09/2025

Abstract

Customer sentiment analysis plays a vital role in understanding audience insights in movies, as online reviews and ratings have become an important source of feedback in the film industry. With the rise of social media platforms, blogs, and review sites, large amounts of content created by users, provide valuable insights into how films are received by various audiences. This research focus on analysing customer sentiment in movie reviews using traditional natural language processing (NLP) and machine learning techniques. Methods such as text preprocessing, tokenisation, and feature extraction through approaches like TF-IDF are implemented to prepare review data for classification. Supervised learning algorithms, including Naïve Bayes, Support Vector Machines and Linear Regression SGD-Based, are used to categorise reviews into positive, negative, or neutral sentiments. This research also looked at the challenges such as handling slang, sarcasm, and domain-specific vocabulary that often shows in informal movie interactions. Findings highlight the potential of sentiment analysis to support decision-making in the film industry, from predicting box office performance to improving recommendation systems and marketing strategies. additionally, analysing customer sentiment in movie reviews provides a meaningful understanding of audience engagement and offers real applications for producers, marketers, and film supporters.

Contents

Declaration.....	1
Abstract.....	2
1 Introduction.....	4
1.1 Background.....	4
1.2 Aims.....	4
1.3 Objectives.....	4
1.4 Research Approach.....	4
1.5 Structure of the Report.....	4
1.6 Ethical, Social, Professional, Legal and Security Considerations.....	4
2 Literature Review	5
3 Practical Research Methodology.....	6
3.1 Approach 1.....	6
4 Analysis of Results	8
4.1 Model Results.....	8
4.3 Comparison of Results with Existing Literature	8
5 Project Evaluation and Reflection.....	9
5.1 Overview of Project.....	9
5.2 Objectives Review.....	9
5.3 Evaluation of Methodology and Results.....	9
5.4 Personal Reflection.....	9
5.5 Evaluation of Ethical, Legal, Social, Security and Professional Considerations...	9
6 Conclusion and Recommendations.....	10
6.1 Conclusion.....	10
6.2 Recommendations.....	10
7 Reference List.....	11
Appendix A. Research Proposal.....	12

1 Introduction

1.1 Background

In the modern digital landscape, the advent and widespread use of social media platforms have profoundly transformed how consumers interact with products and express their opinions. With billions of users actively sharing their thoughts across platforms such as Twitter (now X), Facebook, Reddit, IMDB, Instagram and YouTube, social media has evolved into a dynamic ecosystem of real-time public discourse (Rodríguez-Ibáñez *et al.*, 2022). Within this context, the entertainment industry especially the movie sector has become one of the most discussed and scrutinized domains online. Audiences now play an active role not only in consuming film content but also in shaping its perception through instantaneous reviews, reactions, memes, and hashtags (Xu, Chang, & Jayne, 2022).

Traditionally, movie reviews were limited to film critics and media outlets. However, the democratization of content creation through social media has given every viewer a voice (Amrutiya *et al.*, 2023). These opinions, expressed in text, emojis, images, or videos, have substantial influence. A trending hashtag or viral comment can boost or damage a film's box office performance within hours of release (Choudhury *et al.*, 2021). Consequently, understanding public sentiment has become an essential task for stakeholders in the film industry, including producers, distributors, marketers, and researchers (Yadav *et al.*, 2025).

Given the enormous volume of user-generated content, manually assessing and interpreting this data is impractical. This is where Sentiment Analysis a subfield of Natural Language Processing (NLP) becomes invaluable (Batra *et al.*, 2021). Sentiment analysis enables machines to interpret, classify, and quantify emotions embedded in textual data (Iqbal *et al.*, 2022). It categorizes opinions as positive, negative, or neutral and can extract deeper emotional tones such as anger, joy, or disappointment (Pang *et al.*, 2021). In the context of movie reviews, it offers a scalable way to gauge audience reception and identify patterns or shifts in sentiment over time (Hossain *et al.*, 2022).

Moreover, advances in machine learning (ML) and deep learning have made it possible to train models capable of analyzing complex, context-dependent language (Gowri *et al.*, 2022).

Techniques like Logistic Regression, Support Vector Machines (SVM), Naive Bayes, and neural networks such as LSTM (Long Short-Term Memory) and Transformers can process social media data with impressive accuracy (Durairaj & Chinnalagu, 2021; Gopalakrishnan & Salem, 2020). When integrated with data visualization tools, these models can produce real-time sentiment dashboards, trend charts, and keyword clouds that reveal not only how people feel but also why they feel that way (Bacco *et al.*, 2021).

This research emerges from the need to harness the power of these technologies to create an intelligent system that can automatically analyse sentiment from social media posts related to a specific movie product (Zumma *et al.*, 2022). Beyond classifying sentiment polarity, the goal is to derive insights that can inform content creators and marketers about audience preferences, perception gaps, and potential improvements (Godia & Tiwari, 2024). As social media continues to serve as both a feedback loop and a promotional arena for movies, the ability to interpret its pulse accurately is not just advantageous it is essential (Karimi *et al.*, 2021).

Ultimately, this study contributes to the growing body of work in computational social science, demonstrating how data science can bridge the gap between technology and human behavior (Nguyen *et al.*, 2020). The implications go beyond the movie industry, offering a template for sentiment analysis in broader product review contexts across different sectors (Paranjape, 2022).

1.2 Aims

To create a sentiment analysis model that uses natural language processing (NLP) techniques to categorize product reviews and social media posts as positive, negative, or neutral; to extract important customer sentiments and themes; and to assess how sentiment insights can help improve customer satisfaction, direct product enhancements, and support business marketing campaigns.

1.3 Objectives

To achieve the stated aim, the study will pursue the following specific objectives:

1. To gather and clean a dataset of 50,000 social media posts and product reviews (IMDB), ensuring the data is preprocessed and ready for sentiment analysis (e.g., removing noise, tokenizing text, and handling stopwords).

2. To build and compare at least two sentiment analysis models such as a traditional machine learning model and a deep learning approach, aiming for a sentiment classification accuracy of at least 75%.
3. To implement sentiment analysis models using machine learning and deep learning approach and NLP techniques to classify the polarity (positive, negative, neutral) of social media comments and posts.
4. To evaluate the performance of sentiment classification models based on metrics such as accuracy, precision, recall, F1-score and cross validation.
5. To identify common keywords, themes, and patterns within social media commentary related to the selected movie product.
6. To visualize public sentiment trends and distributions over time using suitable data visualization techniques.

1.4 Research Questions

This research is guided by the following key questions:

7. In what ways can natural language processing be used to analyze customer sentiments in social media and product reviews that can help and assist for product improvement and marketing decisions?
8. What is the overall sentiment distribution (positive, negative, neutral) of social media users regarding the selected movie?
9. Which sentiment analysis model yields the best performance in classifying social media text?
10. What are the most frequently occurring keywords and themes in user reviews on social media platforms?

11. How do sentiment trends evolve before, during, and after the release of a movie, and what insights can be drawn from this evolution?

1.5 Significance of the Study

The significance of this study lies in its ability to demonstrate how sentiment analysis and natural language processing (NLP) techniques can be applied to extract valuable insights from vast volumes of unstructured social media data (Xu, Shu, Yu, & Liu, 2020). In an age where consumer opinions travel faster and farther than ever before, real-time understanding of public sentiment is a competitive advantage especially in the entertainment industry (Rodríguez-Ibáñez *et al.*, 2022).

This study contributes to both academia and industry. Academically, it enriches the body of knowledge in the fields of machine learning, sentiment analysis, and social media analytics by showcasing an applied case study focused on movie product reviews (Iqbal *et al.*, 2022). For practitioners, particularly movie producers, marketers, and data analysts, the study provides a framework for monitoring audience feedback, optimizing marketing strategies, and predicting movie performance based on public reactions (Amrutiya *et al.*, 2023).

Furthermore, by identifying trending keywords, emotional tones, and temporal sentiment patterns, the system developed in this study can serve as a decision-support tool for content development and consumer engagement planning (Hossain *et al.*, 2022). As media consumption becomes increasingly digitized, such tools are vital for aligning creative outputs with audience expectations (Bacco *et al.*, 2021).

1.6 Scope and Limitations

Scope

This study focuses on analysing public sentiment related to a specific movie product (IMDB) by collecting user-generated content from social media platforms, primarily Twitter and Reddit, due to their openness and ease of data access via APIs (Nguyen *et al.*, 2020). The analysis will cover

a specific timeframe: one month before and one month after the release of the selected movie. The primary data type considered is text; multimedia content such as videos or memes will not be analysed (Surya, 2023).

The study will use supervised machine learning and natural language processing methods to classify sentiment and extract thematic insights (Furtado & Singh, 2020). The implementation will be conducted using Python and relevant NLP libraries such as NLTK, Scikit-learn, and possibly deep learning libraries such as TensorFlow or PyTorch (Gopalakrishnan & Salem, 2020).

Limitations

12. The study is limited to text-based data; it does not account for sarcasm, irony, or linguistic nuances that may affect sentiment accuracy (Durairaj & Chinnalagu, 2021).
13. Social media data may contain noise such as spam or irrelevant posts, which could influence the quality of the dataset (Pota *et al.*, 2020).
14. Only a limited number of platforms (e.g., IMDB, Twitter and Reddit) will be used for data collection due to accessibility and API limitations (Karimi *et al.*, 2021).
15. Sentiment classification will be constrained to basic polarities: positive, negative, and neutral. More complex emotional states are outside the scope of this research (Batra *et al.*, 2021).
16. Time constraints may limit the dataset size and the depth of model tuning and evaluation (Yadav *et al.*, 2025).

1.7 Research Approach

This study adopts a quantitative and computational research approach aimed at analysing customer sentiment expressed on social media platforms in relation to a specific movie product. The process begins with data collection from platforms such as IMDB, Twitter and Reddit using publicly available APIs. Relevant posts will be extracted based on movie-related keywords and hashtags within a defined timeframe surrounding the movie's release.

Collected data will undergo preprocessing to remove noise and standardize the text. This involves tokenization, lowercasing, removal of stop-words, and text normalization through stemming or lemmatization. Once cleaned, the data will be classified using supervised machine learning models such as naive bayes, support vector machine or linear regression, These models will be trained to categorize sentiment into positive, negative, or neutral classes.

Model performance will be evaluated using accuracy, precision, recall, F1-score, confusion matrix, regression metrics and cross-validation. Sentiment trends will then be visualized using charts, word clouds, and timelines to identify patterns in public opinion. Finally, the results will be interpreted to derive actionable insights that can inform marketing strategies, content planning, and audience engagement.

The research is grounded in a positivist paradigm, emphasizing objective data analysis and reproducibility. Tools such as Python, Scikit-learn, NLTK, and Matplotlib will be used for model development and data visualization.

1.8 Structure of the Report

This report is structured into five main chapters as outlined below:

- **Chapter One: Introduction**

Provides an overview of the study, including the background, aims, objectives, research questions, significance, and structure of the report.

- **Chapter Two: Literature Review**

Reviews related academic and industry work on sentiment analysis, NLP techniques, machine learning models, and their application in movie product evaluation.

- **Chapter Three: Methodology**

Describes the research design, data collection methods, preprocessing steps, machine learning techniques, tools, and evaluation strategies used in the study.

- **Chapter Four: Results and Analysis**

Presents the experimental results, including model performance metrics, sentiment distribution, visualizations, and thematic interpretations.

- **Chapter Five: Project Evaluation and Reflection**

Skills in data interpretation, sentiment analysis and critical thinking were done by in this research. It strengthen both technical and reflective research skill and improved comprehension despite the slangs and sarcasm.

- **Chapter Five: Conclusion and Recommendations**

Summarizes key findings, discusses implications, acknowledges limitations, and provides recommendations for future research and practical applications.

1.9 Ethical, Social, Professional, Legal and Security Considerations

1.9.1 Professional Considerations

This research will be conducted in line with the standards of professional conduct expected in data science, computer science, and academic research (Ganesh *et al.*, 2021). Proper documentation of methods, transparency in reporting results, and reproducibility of findings will be maintained throughout. All models, tools, and findings will be shared with full disclosure of limitations to avoid overclaiming accuracy or utility.

The researcher will also be mindful of algorithmic bias, ensuring that the sentiment analysis models are trained and tested using balanced data to prevent skewed interpretations (Hassan *et al.*, 2023). Any tools developed will be used responsibly and not for unethical surveillance, profiling, or discrimination.

1.9.2 Ethical Considerations

Although social media platforms provide publicly accessible content, ethical concerns still arise regarding the privacy and autonomy of individuals (Phan & Do, 2020). This study ensures that no personally identifiable information (PII) is collected, stored, or used. Data will be anonymized, and users' handles, profile details, and metadata will be excluded from analysis. Moreover, the

research will refrain from manipulating or influencing any ongoing online discussions (Zumma *et al.*, 2022). All analyses will be observational and retrospective.

To further uphold ethical standards, data will be used strictly for academic purposes and will not be exploited for commercial gain. The research will also ensure fairness by evaluating sentiment without bias toward any user demographic, culture, or political group (Smitha *et al.*, 2021).

1.9.3 Legal Considerations

The research complies with the terms of service and data usage policies of all social media platforms from which data is collected, such as Twitter's Developer Policy and Reddit's API terms (Kowalczyk, n.d.). The study also aligns with relevant data protection laws, including the General Data Protection Regulation (GDPR) and Nigeria Data Protection Regulation (NDPR), where applicable. Only data explicitly permitted by these platforms for academic and research purposes will be accessed, ensuring legal compliance throughout the project.

No copyrighted content will be reproduced in the report, and all third-party tools and datasets used will be properly cited in accordance with academic integrity standards (Aleqabie *et al.*, 2020).

1.9.4 Social Considerations

Social media sentiment analysis can shape public perception and influence behavior. Therefore, responsible reporting of results is crucial (Rodríguez-Ibáñez *et al.*, 2022). This study avoids drawing misleading conclusions from limited or potentially biased datasets. The researcher will also be cautious not to generalize sentiments beyond the scope of the collected data or ignore the cultural and contextual nuances embedded in language (Xu, Chang, & Jayne, 2022).

2 Literature Review

2.1 Introduction and Sentiment Analysis Landscape

In today's digital era, social media platforms like Twitter, IMDb, and YouTube have become prominent sources of real-time, user-generated content, offering valuable insights into public opinion. This explosion of unstructured textual data has led to the rise of sentiment analysis, a Natural Language Processing (NLP) technique used to determine whether a text expresses positive, negative, or neutral sentiment (Hossain *et al.*, 2022; Aleqabie *et al.*, 2020). Its applications span industries such as e-commerce, politics, and particularly entertainment, where sentiment analysis helps gauge audience responses to films, trailers, and casting announcements (Choudhury, Mohanty, & Jagadev, 2021).

Social platforms present challenges for traditional models due to their informal language, slang, and sarcasm (Baseiso & Alzahrani, 2020). Nevertheless, they remain essential for understanding diverse consumer sentiments and shaping marketing or content strategies (Nguyen, Vu, & Nguyen, 2020). Studies show that user feedback on IMDb and Twitter can influence movie success, acting as early indicators of public perception (Zumma *et al.*, 2022; Singh, Jakhar, & Pandey, 2021).

2.2 Sentiment Analysis in Movie Reviews

In the modern film industry, sentiment analysis has become a crucial tool for understanding how audiences feel about movies, both before and after their release. By analysing large volumes of audience-generated text—such as comments, reviews, and social media posts—researchers and industry professionals can gain valuable insights into public opinion, predict box office performance, and make informed marketing and production decisions.

Online platforms provide different types of sentiment-rich data. IMDb offers well-structured, long-form reviews with star ratings, which allow for in-depth analysis of opinions. In contrast, platforms like Twitter capture brief, real-time reactions to trailers, casting announcements, or the film itself, making it possible to track audience sentiment trends almost instantly (Paranjape, 2022; Qaisar, 2020). YouTube comment sections also play a significant role, especially for analysing

audience responses to official trailers, review videos, or behind-the-scenes clips. Together, these sources provide a comprehensive picture of audience engagement, ranging from quick emotional responses to detailed reflections.

The relationship between sentiment and a film's commercial success is well documented. Positive sentiment before a film's release—such as praise for a trailer or early critic reviews—can increase anticipation, encourage ticket pre-sales, and boost opening-weekend revenue. On the other hand, negative early reactions can discourage potential viewers, lower attendance, and force studios to adapt their marketing strategies. In some cases, widespread online criticism has even led to changes in promotional materials or alterations to the final product (Pang et al., 2021).

Historically, sentiment analysis in movie reviews began with traditional machine learning (ML) approaches such as Naive Bayes, Support Vector Machines (SVM), and Logistic Regression. These models rely on manually designed features, using methods like Bag of Words (BoW), Term Frequency–Inverse Document Frequency (TF-IDF), and N-grams to convert text into numerical form (Furtado & Singh, 2020; Smitha & Bharath, 2020). While these techniques are relatively simple and computationally efficient, they struggle to capture context, sarcasm, or complex sentence structures. For example, a sentence like “I didn’t hate the movie” might be wrongly classified as negative if the model only focuses on the word “hate” without understanding the negation.

With the advancement of deep learning (DL), sentiment analysis has become far more accurate and context-aware. Long Short-Term Memory (LSTM) networks and Bidirectional LSTMs (BiLSTM) can track the meaning of words across an entire review, handling long-range dependencies and changes in tone throughout a sentence or paragraph (Abdelgwad et al., 2021). This makes them especially effective for longer, more detailed movie reviews where the sentiment may shift over time.

The most significant progress in recent years has come from transformer-based models, particularly BERT (Bidirectional Encoder Representations from Transformers). Unlike earlier models, BERT processes text in both directions at the same time—looking at the words before and after a target word—to fully understand its meaning in context. This approach allows BERT to detect subtle expressions, sarcasm, and even mixed sentiments more effectively. Studies show that

BERT-based models often achieve accuracy levels above 90% in movie review sentiment classification, outperforming both traditional ML and earlier DL models (Batra et al., 2021).

Beyond academic research, sentiment analysis has practical applications in the entertainment industry. Streaming platforms like Netflix and Amazon Prime Video use sentiment data to improve recommendation systems, matching viewers with content that aligns with their tastes. Film studios apply sentiment insights to predict audience reception to casting decisions, plot themes, or even marketing campaigns. Real-time analysis of social media reactions during a film's release can help studios make quick adjustments, such as changing promotional content or targeting specific audience segments.

2.3 Machine Learning Techniques in Sentiment Analysis

Traditional machine learning (ML) techniques, such as Naive Bayes, Support Vector Machines (SVM), and Logistic Regression, have long been applied in sentiment analysis due to their simplicity, interpretability, and relatively low computational requirements (Joulin et al., 2016; Choudhury, Mohanty, & Jagadev, 2021). These algorithms rely on statistical learning principles and typically use engineered textual features, such as Bag of Words (BoW) or Term Frequency–Inverse Document Frequency (TF-IDF), for classification.

While they perform adequately in structured and formal text environments, their effectiveness tends to decline in contexts involving informal language, sarcasm, or rapidly evolving slang, which are common in social media communications. Nonetheless, they remain valuable in low-resource settings, where computational constraints or limited training data make complex models impractical (Kowalczyk, n.d.).

2.4 Deep Learning Techniques in Sentiment Analysis

Deep learning (DL) approaches, such as Convolutional Neural Networks (CNN), Long Short-Term Memory networks (LSTM), and transformer-based architectures like Bidirectional Encoder Representations from Transformers (BERT), have significantly advanced sentiment analysis performance. Unlike traditional ML models, DL methods can automatically learn hierarchical and contextual language representations directly from raw text, reducing reliance on manual feature

engineering.

CNNs excel at capturing local semantic features and patterns within sentences (Surya, 2023), while LSTM and Bidirectional LSTM (BiLSTM) networks are adept at modelling long-term dependencies and sequential structures in text, which is particularly useful in understanding the sentiment of longer reviews (Danyal et al., 2023; Gopalakrishnan & Salem, 2020). Transformer-based models, especially BERT, have set new performance benchmarks in sentiment classification, achieving accuracy rates exceeding 90% on movie review datasets (Ganesh et al., 2021; Yadav et al., 2025). Comparative studies have consistently reported that BERT-based architectures outperform both traditional ML and earlier deep learning approaches (Durairaj & Chinnalagu, 2021; Phan & Do, 2020). However, these improvements come at the cost of increased computational power requirements, making them more suitable for large-scale or enterprise-level sentiment analysis applications.

2.5 Data Acquisition and Pre-processing

The quality and reliability of any sentiment analysis system depend significantly on how well the data is acquired and prepared before model training. Social media and online platforms present a rich yet complex source of textual data. However, raw text data from these sources is often noisy, inconsistent, and unstructured. As such, proper acquisition and pre-processing are essential steps in building a robust sentiment analysis pipeline, particularly for movie product reviews (Surya, 2023; Zumma *et al.*, 2022).

2.5.1 Data Sources and Collection Methods

Sentiment analysis for movie reviews typically involves collecting data from a variety of online platforms where users share their opinions and feedback. Common data sources include:

- **IMDb (Internet Movie Database):** Provides user reviews along with star ratings. Many researchers use the IMDb 50K review dataset, which includes an equal distribution of positive and negative reviews, making it ideal for sentiment classification experiments (Maas *et al.*, 2011).
- **Twitter:** Offers real-time and spontaneous reactions to movie trailers, announcements, and releases. Data is typically collected using Twitter APIs or third-party tools like

Tweeepy, filtering tweets by hashtags, keywords, or user mentions (Nguyen, Vu, & Nguyen, 2020).

- **YouTube:** Comment sections under trailers and review videos are another rich source of sentiment-laden text, often reflecting unfiltered public opinion (Aleqabie *et al.*, 2020).
- **App Store and Google Play Reviews:** Especially relevant in studies involving streaming platforms like Netflix and Amazon Prime, where users leave app feedback that includes reactions to content (Pota *et al.*, 2020).
- **Web scraping and APIs:** Custom scripts are frequently written using tools such as BeautifulSoup, Selenium, or platform-specific APIs to collect review data, with ethical considerations and terms of service compliance in mind (Smitha & Bharath, 2020).

2.5.2 Text Cleaning and Normalization

Once data is collected, it must undergo rigorous pre-processing to convert raw text into a format suitable for analysis. The steps generally include:

- **Text Cleaning:** Removing unwanted elements such as HTML tags, URLs, emojis, numbers, special characters, and excessive white spaces (Gopalakrishnan & Salem, 2020).
- **Tokenization:** The process of splitting text into individual words, phrases, or symbols (tokens). Tokenization is fundamental for further steps like stemming, lemmatization, and vectorization (Andrade-Segarra & Leon-Paredes, 2021).
- **Lowercasing:** Converting all characters to lowercase to maintain uniformity and reduce redundancy.
- **Stop-word Removal:** Eliminating common words such as "is", "the", "and", which do not carry meaningful sentiment but inflate the feature space (TextBlob, as cited by Aleqabie *et al.*, 2020).
- **Stemming and Lemmatization:**
 - Stemming reduces words to their root form (e.g., “watching” → “watch”) (NLTK: Hossain *et al.*, 2022).

- Lemmatization uses vocabulary and grammar analysis to return the base or dictionary form of a word (e.g., “was” → “be”) (SpaCy: Pota *et al.*, 2020).

2.5.3 Handling Class Imbalance

In real-world datasets, particularly those scraped from social platforms, there is often a class imbalance—for instance, an overrepresentation of positive reviews compared to negative ones. Class imbalance can skew the training process, causing models to become biased toward the majority class (Danyal *et al.*, 2023).

One widely adopted method to address this issue is the Synthetic Minority Over-sampling Technique (SMOTE). Other techniques include:

- Random Over-sampling or Under-sampling
- Ensemble methods with cost-sensitive learning
- Stratified cross-validation during model evaluation

These approaches are essential to mitigate bias and improve fairness in model predictions (Choudhury, Mohanty, & Jagadev, 2021).

2.6 Feature Extraction and Representation

Once textual data has been collected and pre-processed, the next critical step in sentiment analysis is converting that data into numerical representations that machine learning or deep learning models can understand. This process is known as feature extraction or text vectorization. The quality of these features greatly influences the performance of sentiment classification models, particularly when dealing with complex and opinion-rich content such as movie reviews (Furtado & Singh, 2020; Prince Praveen, Goud, & Parmar, 2020).

2.6.1 Bag of Words (BoW)

The Bag of Words model is one of the earliest and simplest techniques for feature extraction. It represents text by counting the frequency of words within a document, disregarding grammar and word order but retaining multiplicity. Each document is represented as a fixed-length vector, where each dimension corresponds to a word from the entire vocabulary (Smitha & Bharath, 2020).

While BoW is easy to implement and often effective in basic classification tasks, it has significant limitations:

- It ignores the context and semantics of words.
- It creates very large and sparse feature vectors.
- It treats all words as equally important, which may not be ideal in sentiment-rich texts (Joulin *et al.*, 2016).

Despite these limitations, BoW has served as a solid baseline in many earlier sentiment analysis studies, including those focusing on IMDb movie reviews (Maas *et al.*, 2011).

2.6.2 Term Frequency–Inverse Document Frequency (TF-IDF)

The TF-IDF method enhances the BoW approach by considering not only the frequency of a word in a document (term frequency), but also how unique that word is across all documents (inverse document frequency). Words that appear frequently in one document but rarely across the entire corpus are given more weight (Arora, 2020).

TF-IDF vectors help mitigate the problem of common but sentiment-neutral words dominating the feature space. This method is particularly useful when classifying user reviews where unique, opinionated language can provide stronger signals than frequently occurring general terms (Danyal *et al.*, 2023).

Several studies in the reviewed literature have used TF-IDF to train traditional ML models like SVM, Logistic Regression, and Random Forests on movie review datasets, reporting improved performance over BoW (Bacco *et al.*, 2021; Singh, Jakhar, & Pandey, 2021).

2.6.3 N-grams

N-gram models are a simple yet powerful extension of BoW that consider combinations of n consecutive words. For instance:

- Unigrams: single words
- Bigrams: two-word sequences (e.g., “very good”, “not bad”)

- Trigrams: three-word phrases

N-grams help preserve some degree of word order and local context, which is crucial for capturing sentiment nuances such as negation (“not good” vs “good”) and emphasis (Zumma *et al.*, 2022). However, they can significantly increase the dimensionality of the feature space and often require pruning to remove rare or irrelevant phrases (Qaisar, 2020).

2.6.4 Word Embeddings (Word2Vec, GloVe)

To overcome the limitations of sparse and context-unaware vectorization methods like BoW and TF-IDF, word embeddings were introduced. These techniques map words into dense, low-dimensional vectors where semantically similar words are located near each other in the vector space.

- Word2Vec learns word associations from large text corpora using either the Continuous Bag of Words (CBOW) or Skip-gram model. It captures semantic relationships such as analogies (e.g., "king" – "man" + "woman" $\approx$ "queen") (Joulin *et al.*, 2016).
- GloVe (Global Vectors for Word Representation) is another unsupervised learning algorithm that uses global word co-occurrence statistics to learn vector representations (Batra *et al.*, 2021).

Word embeddings enable sentiment models to better understand relationships and sentiment polarity by embedding contextual and semantic meaning into each word’s representation. They are widely used in deep learning models such as LSTM and BiLSTM architectures for movie review classification (Gopalakrishnan & Salem, 2020).

2.6.5 Contextualized Word Embeddings (BERT, USE)

Traditional embeddings assign a single vector to each word, regardless of its context. Contextualized embeddings, on the other hand, generate dynamic word representations based on surrounding words in a sentence. This advancement greatly enhances sentiment analysis, especially in domains where language is rich with nuance and ambiguity, such as film reviews (Karimi, Rossi, & Prati, 2021).

- **BERT (Bidirectional Encoder Representations from Transformers):** BERT processes text bidirectionally and captures both left and right context simultaneously. Fine-tuned BERT models have consistently achieved state-of-the-art performance in sentiment classification tasks (Ganesh *et al.*, 2021; Hassan, Khare, & Paik, 2023).
- **Universal Sentence Encoder (USE):** Developed by Google, USE encodes entire sentences or paragraphs into fixed-length vectors, making it ideal for sentence-level sentiment analysis. It has been used in review summarization and transfer learning across domains (Xu *et al.*, 2020).

Contextualized embeddings like BERT have shown strong results in recent studies. For instance, experiments using BERT on IMDb datasets have reported accuracy rates above 90%, outperforming traditional and static embedding methods by a significant margin (Geetha & Renuka, 2021).

2.7 Evaluation Metrics for Sentiment Classification

Evaluating the performance of sentiment analysis models is crucial for determining their reliability, generalisability, and practical value. The effectiveness of a classifier is not judged solely by how often it makes correct predictions, but also by how well it balances between detecting positive and negative sentiments, especially in datasets with class imbalances (Choudhury, Mohanty, & Jagadev, 2021).

2.7.1 Accuracy

Accuracy is the most intuitive and widely used metric in classification tasks. It represents the proportion of correctly predicted instances among the total number of predictions. Although useful, accuracy alone can be misleading, especially when the dataset is imbalanced. For example, if 80% of the reviews are positive, a model that always predicts "positive" will achieve high accuracy but poor performance on negative sentiment detection (Surya, 2023).

2.7.2 Precision, Recall, and F1-score

To address the limitations of accuracy, precision, recall, and the F1-score are commonly used to provide a more detailed evaluation.

F1-score is particularly important when the cost of misclassification varies between sentiment classes, such as wrongly classifying a critical negative movie review as positive (Hossain *et al.*, 2022; Danyal *et al.*, 2023).

2.7.3 ROC-AUC (Receiver Operating Characteristic – Area Under Curve)

ROC-AUC is a threshold-independent metric that assesses a model's ability to distinguish between classes. AUC (Area Under the Curve) summarizes the ROC curve in a single value between 0 and 1. A value closer to 1 indicates excellent model performance (Bacco *et al.*, 2021).

In sentiment analysis, especially when deploying models for real-time feedback systems, AUC can help assess how well a model differentiates between strong and weak sentiments, regardless of the threshold set for classification (Akhand *et al.*, 2021).

2.7.4 Confusion Matrix

The confusion matrix is a visual representation of classification outcomes. It tabulates the actual vs. predicted sentiment classes and provides detailed insight into model errors. This matrix helps in identifying whether a model is biased towards over-predicting a particular class and whether specific types of misclassifications are frequent (Zumma *et al.*, 2022; Hassan, Khare, & Paik, 2023).

2.7.5 Model Benchmarking Methods

To ensure the credibility of evaluation metrics, cross-validation and benchmarking against baseline models are essential:

Benchmarking advanced models such as BERT or BiLSTM against traditional models like TF-IDF with Logistic Regression is a common practice in recent studies (Phan & Do, 2020; Pang *et al.*, 2021). K-fold cross-validation remains one of the most reliable methods to evaluate model consistency and generalisability.

2.8 Tools and Frameworks for Sentiment Analysis

2.8.1 Python Libraries for Machine Learning and Deep Learning

Python has become the dominant language for sentiment analysis due to its robust ecosystem of libraries and ease of use. Some of the most widely adopted libraries include:

- **Scikit-learn:** Provides efficient implementations of classical algorithms such as Naive Bayes, SVM, and Logistic Regression (Smitha & Bharath, 2020).
- **TensorFlow:** Supports scalable deep learning models like CNNs and LSTMs, widely used for review classification (Gopalakrishnan & Salem, 2020).
- **Keras:** A high-level API for deep learning that simplifies model development and is ideal for rapid prototyping (Paranjape, 2022).
- **PyTorch:** Offers dynamic computation graphs and is commonly used for transformer-based models such as BERT (Ganesh *et al.*, 2021).

2.8.2 NLP Toolkits and APIs

- **NLTK:** Offers tools for tokenization, stemming, and lemmatization, and remains popular in academic settings (Maas *et al.*, 2011).
- **TextBlob:** Good for quick polarity scoring but often lacks domain specificity (Aleqabie *et al.*, 2020).
- **SpaCy:** Efficient for large-scale linguistic processing, including named entity recognition and dependency parsing (Pota *et al.*, 2020).
- **VADER:** Especially effective for analyzing short, informal texts such as tweets and YouTube comments (Nguyen, Vu, & Nguyen, 2020).
- **Hugging Face Transformers:** A modern standard that provides access to thousands of pretrained models, enabling fine-tuning for domain-specific sentiment tasks (Hassan, Khare, & Paik, 2023).

2.9 Review of Selected Research Studies

This section presents a structured analysis of thirteen peer-reviewed studies focused on sentiment analysis, particularly in the domain of movie and product reviews from social media platforms. These studies differ in terms of the algorithms employed, datasets used, and evaluation techniques applied. Together, they offer insight into current trends, methodological challenges, and areas for future development.

No.	Title	Authors	Year	Focus	Dataset	Accuracy/Result
1	Sentiment Analysis of Consumer Reviews Using Deep Learning	Iqbal <i>et al.</i>	2022	LSTM variations for sentiment prediction	Various textual datasets	Improved results, lower complexity
2	Transformer-based Contextual Model for Sentiment Analysis of Customer Reviews	Durairaj & Chinnalagu	2021	Fine-tuned BERT vs others	Twitter, IMDB, Yelp, Amazon	BERT outperformed others
3	Sentiment Analysis of Movie Reviews Using Machine Learning	Yadav <i>et al.</i>	2025	ML vs DL models on IMDB	IMDb 50K	BERT: 94.2% accuracy

4	ML-Based Sentiment Analysis of Movie Reviews	Amrutiya <i>et al.</i>	2023	ML algorithm comparison	Movie reviews (general)	SVM+IG: 85.65%, RF: 82.1%
5	Sentiment Analysis of IMDb Movie Reviews Using LSTM	Qaisar	2022	LSTM for classification	IMDb 50K	Accuracy: 89.9%
6	Sentiment Classification of Product Reviews	Godia & Tiwari	2024	TF-IDF + SMOTE + ML models	Flipkart reviews	Best model via hyperparameter tuning
7	ML for Sentiment Analysis of Movies Using IMDb	Rathor & Prakash	2024	IMDB + ML framework	IMDB	SVM: 90.3% accuracy
8	Neural Network Performance on IMDb Reviews	Haque <i>et al.</i>	2019	CNN vs LSTM vs LSTM-CNN	IMDb	CNN: F1-score 91%
9	Review on Sentiment Analysis from Social Media	Rodríguez-Ibáñez <i>et al.</i>	2022	Trends and applications in social platforms	Social media (general)	Identifies gaps and domain trends

10	DL vs ML for OTT App Sentiment	Arpita <i>et al.</i>	2024	User review analysis on OTT	Play Store & App Store (56K reviews)	BiLSTM: 92%, LR: 66.6%
11	Systematic Review of Social Media Sentiment Analysis	Xu <i>et al.</i>	2022	Challenges, trends, evaluations	Various platforms	Identifies language/data/evaluation issues
12	Improved Sentiment Analysis using Naive Bayes	Gowri <i>et al.</i>	2022	Naive Bayes sentiment model	Jai Bhim reviews	Enhanced NB performance
13	Review on Movie Recommendation Systems	Smitha <i>et al.</i>	2021	Hybrid recommender systems	Movie datasets	Supports sentiment-based recommendation

Key Observations from Reviewed Studies

1. Model Choice and Algorithm Selection

The reviewed literature demonstrates a significant shift from traditional machine learning to deep learning and transformer-based approaches. Classical models such as Naive Bayes and SVM remain relevant but are gradually being outperformed by modern architectures like LSTM and BERT.

For example, Iqbal *et al.* (2022) proposed three LSTM variations that demonstrated efficient sentiment classification with lower computational costs. Similarly, Qaisar (2022) applied a standard LSTM model on IMDb reviews and achieved an accuracy of 89.9%, supporting the effectiveness of sequence-based learning.

More advanced models like BERT, as employed by Durairaj and Chinnalagu (2021) and Yadav *et al.* (2025), were shown to outperform traditional ML and hybrid models in multiple datasets, with Yadav *et al.* achieving a notable 94.2% accuracy using BERT on IMDb reviews. These results underline the growing dominance of transformer-based architectures in sentiment classification.

2. Dataset Quality and Domain Alignment

IMDb was the most frequently used dataset due to its balanced structure and availability of labelled sentiment data (Yadav *et al.*, 2025; Qaisar, 2022; Haque *et al.*, 2019). Rathor and Prakash (2024) also relied on IMDb, testing classifiers like Naive Bayes and SVM, and reported an SVM accuracy of 90.3%.

Meanwhile, Arpita *et al.* (2024) collected over 56,000 user reviews from the Play Store and App Store, focusing on OTT platforms. Their results showed that BiLSTM achieved 92% accuracy, indicating that domain-specific data with contextual features can benefit from advanced DL models.

3. Feature Engineering and Pre-processing Techniques

Earlier studies (Amrutiya *et al.*, 2023; Godia & Tiwari, 2024) relied on traditional techniques like TF-IDF and Bag of Words. Godia and Tiwari introduced SMOTE to address class imbalance and improved classifier accuracy through hyperparameter tuning.

In contrast, recent approaches leveraged pre-trained word embeddings and contextual representations. For instance, Durairaj and Chinnalagu (2021) and Yadav *et al.* (2025) used

BERT embeddings to significantly enhance performance without extensive manual feature engineering.

4. Evaluation Methods and Reporting Metrics

Evaluation metrics such as accuracy, F1-score, precision, and recall were widely reported. CNN achieved the highest F1-score (91%) in Haque *et al.*'s (2019) comparative analysis of neural networks, while the hybrid BiLSTM-FastText model used by Arpita *et al.* (2024) also demonstrated strong performance.

Xu *et al.* (2022) and Rodríguez-Ibáñez *et al.* (2022) both discussed the importance of choosing appropriate evaluation metrics, especially when dealing with real-time, short-form, and multilingual social media texts. Their systematic reviews identified gaps in model benchmarking and highlighted the need for standardized evaluation protocols.

Grouped Discussion by Theme

A. Movie-Specific Studies

Several studies focused exclusively on movie reviews particularly those using IMDb datasets (Yadav *et al.*, 2025; Qaisar, 2022; Haque *et al.*, 2019; Rathor & Prakash, 2024). These studies confirmed the value of structured, domain-specific corpora and demonstrated that sentiment polarity is strongly linked to linguistic and contextual cues that are well captured by deep learning models.

B. Model Benchmarking

Multiple studies conducted comparative analyses. Durairaj and Chinnalagu (2021) compared BERT to FastText, SVM, and BiLSTM, while Arpita *et al.* (2024) benchmarked DL models against traditional ML classifiers. The consensus was clear: transformer-based models consistently delivered superior results.

C. Social Media and Review Diversity

Studies by Xu *et al.* (2022) and Rodríguez-Ibáñez *et al.* (2022) shifted focus toward broader sentiment analysis challenges such as noisy data, short text length, sarcasm detection, and multilingual limitations all of which are prominent on platforms like Twitter, YouTube, and Facebook.

D. Rule-Based + ML Hybrids

Gowri *et al.* (2022) combined lexicon methods with Naive Bayes for classifying movie reviews from social media, achieving improved sentiment classification without using deep models. This reflects ongoing relevance of lightweight methods in low-resource environments.

2.10 Challenges and Limitations in Sentiment Analysis

Despite significant advancements in sentiment analysis, particularly with the rise of deep learning and transformer-based models, several inherent challenges continue to affect the accuracy, generalizability, and fairness of sentiment classification. These limitations become especially pronounced when analyzing movie product reviews on social media, where content is user-generated, informal, and context-sensitive (Hossain *et al.*, 2022; Batra *et al.*, 2021).

2.10.1. Sarcasm, Slang, and Negation Handling

One of the major limitations in sentiment analysis is the inability of many models especially traditional ones to correctly interpret sarcasm and irony. For instance, the sentence “Wow, that was the best waste of two hours ever!” contains positive words like “best” but expresses a negative sentiment due to sarcasm. Even modern deep learning models may struggle to detect such cues unless they are trained on sufficiently annotated sarcastic corpora (Choudhury, Mohanty, & Jagadev, 2021).

In addition, slang terms and internet-specific expressions (e.g., “meh”, “lit”, “LMAO”, “goat”) evolve rapidly, making it hard for lexicon-based or pretrained models to remain up to date. These informal terms can significantly alter the intended sentiment of a sentence (Zumma *et al.*, 2022).

Negation further complicates sentiment classification. Sentences like “Not bad” or “It’s not the worst I’ve seen” often express mild positivity, which can be misclassified if models fail to detect the modifying role of “not” (Danyal *et al.*, 2023).

2.10.2. Domain Specificity and Transferability

Models trained on movie reviews may perform well in that domain but fail to generalize when applied to other product categories or contexts (e.g., hotel reviews or e-commerce feedback). This issue, known as domain dependency, limits the transferability of trained models across different review types (Prince Praveen, Goud, & Parmar, 2020).

Moreover, sentiment polarity in one domain may not reflect the same intensity or meaning in another. A term like “predictable” may be negative in movie reviews but neutral in financial forecasting. Without domain adaptation or fine-tuning, such models risk reduced accuracy outside their training context (Phan & Do, 2020).

2.10.3. Multilingual and Cross-Cultural Challenges

Most sentiment analysis models are developed and benchmarked using English-language datasets such as IMDb or Amazon reviews. This language bias creates challenges in multilingual or non-English settings where data may be scarce or not well-annotated (Abdelgwad *et al.*, 2021; Akhand *et al.*, 2021).

Additionally, cultural variation affects sentiment expression. For example, phrases or idioms common in one culture may not translate sentiment equivalently in another. This necessitates careful language-specific preprocessing, and in many cases, separate models or multilingual embeddings such as XLM-RoBERTa (Nguyen, Vu, & Nguyen, 2020).

Translating non-English texts into English for analysis (a common workaround) introduces further errors due to misinterpretation or loss of sentiment-laden expressions (Pota *et al.*, 2020).

2.10.4. Computational Cost and Resource Requirements

Advanced models like BERT and GPT-based transformers offer high accuracy but come with substantial computational demands. They require powerful GPUs, large memory capacity, and longer training times, which may not be feasible for researchers or organizations with limited resources (Ganesh *et al.*, 2021; Hassan, Khare, & Paik, 2023).

In real-time systems (e.g., live social media sentiment tracking), inference speed becomes critical. Lightweight models like Naive Bayes or logistic regression are still preferred in time-sensitive or embedded applications, despite their lower performance (Smitha & Bharath, 2020).

Furthermore, fine-tuning large models for specific domains or languages adds to both cost and complexity, making maintenance and scalability challenging (Karimi, Rossi, & Prati, 2021).

2.11 Research Gaps and Future Opportunities

While significant progress has been made in the field of sentiment analysis, especially within the domain of movie product reviews on social media, several research gaps remain. Addressing these gaps could enhance the accuracy, adaptability, and real-world utility of sentiment analysis systems. This section outlines four key areas where future research can contribute meaningfully (Surya, 2023; Geetha & Renuka, 2021).

2.11.1. Real-Time Sentiment Monitoring

Most existing sentiment analysis models are designed for batch processing and are not optimized for real-time execution. Yet, for platforms like Twitter or YouTube, real-time sentiment tracking can provide actionable insights for film producers, marketing teams, and streaming services during or immediately after a movie release (Xu *et al.*, 2020).

2.11.2. Cross-Platform Review Mining

Many current studies focus on a single platform, such as IMDb, Twitter, or YouTube, which limits the generalizability of insights. However, users often express varied opinions across multiple platforms ranging from short, impulsive tweets to long-form YouTube comments and formal IMDb reviews (Nguyen, Vu, & Nguyen, 2020).

Research in this area would require:

- Developing platform-agnostic models,
- Addressing inconsistencies in vocabulary and tone,
- Building pipelines to manage heterogeneous data formats and APIs (Furtado & Singh, 2020).

2.11.3. Multimodal Sentiment Analysis (Text + Visual/Audio)

Current sentiment analysis systems largely focus on textual data, yet user sentiment is also expressed through images, emojis, videos, and speech especially on platforms like TikTok, YouTube, and Instagram. Combining natural language processing (NLP) with computer vision and audio analysis presents an exciting frontier for richer sentiment interpretation (Malki, 2023).

Future work can explore:

- Multimodal neural architectures that fuse text, image, and voice cues,

- Emotion detection from facial expressions or vocal tone,
- Aligning multimodal inputs for cohesive sentiment prediction (Choudhury, Mohanty, & Jagadev, 2021).

Such systems could vastly improve the analysis of video-based movie reviews or reaction content.

2.11.4. Personalization and Context-Aware Sentiment Models

Most models treat sentiment as objective and uniform across users, overlooking individual user preferences, history, and demographics. However, sentiment is often contextual and subjective. For instance, a horror movie might elicit positive sentiment from genre fans but negative responses from others (Batra *et al.*, 2021).

Emerging research can explore:

- User-specific sentiment modeling using preference profiles or viewing history,
- Incorporating temporal context (e.g., mood changes over time),
- Adaptive models that personalize sentiment interpretation based on past behaviors or user demographics (Karimi, Rossi, & Prati, 2021; Andrade-Segarra & Leon-Paredes, 2021).

These innovations would enable more human-like, empathetic sentiment systems, ideal for recommender engines and personalized marketing.

3 Practical Research Methodology

3.1 Research Approach

Sentiment analysis represents a rapidly evolving domain within Natural Language Processing (NLP) that aims to computationally determine the affective orientation of text. This research adopts a quantitative, computational methodology that leverages classical machine learning algorithms for the classification of movie reviews into positive and negative sentiments. The choice of methodology was influenced by both the research objectives and practical considerations such as interpretability, efficiency, and reproducibility.

“This chapter presents the research methodology adopted in this study. The methodology outlines the structured steps followed to conduct and complete the project, ensuring that the results are systematically obtained and aligned with the overall aim. It details the dataset, preprocessing methods, feature extraction, model development, and evaluation procedures.”

The decision to apply machine learning models instead of deep learning architectures such as Bi-LSTM or transformers was deliberate. Although deep learning models often achieve higher accuracies, they require significantly more computational resources, hyperparameter tuning, and large-scale labeled data (Yıldırım & Santur, 2024). In contrast, traditional machine learning models such as Naive Bayes, Support Vector Machines (SVM), and Linear Regression with stochastic gradient descent (SGD) are computationally efficient, interpretable, and still competitive for text classification tasks (Bhardwaj et al., 2023). Additionally, this methodological decision aligns with the project’s scope, which emphasizes explainability and accessibility over state-of-the-art performance.

Another important methodological consideration is ethical responsibility. Sentiment analysis inherently deals with human-generated text, which may include biased or offensive expressions. Consequently, the models must be trained and evaluated with awareness of potential biases in the dataset. For example, IMDB reviews might reflect cultural subjectivity in film evaluation, sarcasm, or linguistic diversity. While this research does not focus on bias mitigation techniques, recognizing such challenges is an essential methodological step, especially as machine learning applications expand into real-world contexts.

This research therefore follows a multi-stage methodology that includes:

1. Dataset selection and description.
2. Text preprocessing and cleaning.
3. Label encoding.
4. Data ingestion and partitioning.
5. Feature extraction using TF-IDF.
6. Model development and training (Naive Bayes, SVM, Linear Regression).
7. Model evaluation using classification and regression metrics.
8. Final model selection.

This structured approach ensures methodological robustness and provides a clear path from raw data to interpretable results, laying the groundwork for Chapter 4, which will present evaluation outcomes.

3.1.1 Creating the Dataset

3.1.2 Dataset Selection and Description

The IMDB movie reviews dataset was chosen as the basis for this study. It is among the most widely used datasets in sentiment analysis research, serving as a benchmark for testing algorithms in both academic and industrial settings. The dataset contains 50,000 movie reviews, evenly distributed between positive and negative sentiment classes. This balance ensures that models trained on the dataset are not inherently biased toward one sentiment category.

The dataset has been referenced in several studies (Tomal, 2024; Bhardwaj et al., 2023) as a reliable benchmark. Its composition makes it ideal for training classifiers because it includes a wide range of linguistic styles, from casual reviews with slang to more formal critique-oriented reviews. This diversity increases the difficulty of classification while simultaneously making it more realistic, as real-world textual data is often noisy and unstructured.

The dataset is formatted as a CSV file with two columns:

- review – the text of the movie review.

- sentiment – the label, either “positive” or “negative.”

This simplicity allows for straightforward preprocessing, feature extraction, and model training.

Despite its strengths, the IMDB dataset is not without limitations. Reviews often contain sarcasm, which is challenging for machine learning algorithms to detect because the literal wording of the review may not reflect the underlying sentiment. Furthermore, writing styles vary significantly; for example, some reviews may be verbose with nuanced language, while others may be short and direct. Finally, while the dataset is balanced in terms of sentiment, it may not be representative of all movie genres or cultural perspectives, introducing potential biases.

Nevertheless, given its widespread use and balance, the IMDB dataset remains one of the most practical and effective resources for sentiment classification, providing both a benchmark for comparison with other studies (Charitha et al., 2023) and sufficient complexity to evaluate multiple algorithms.

3.1.3 Text Cleaning and Preprocessing

Raw text data cannot be directly input into machine learning models. Therefore, a text preprocessing pipeline was applied to transform the dataset into a structured format suitable for analysis.

1. **HTML Tag Removal:** Reviews often include embedded HTML tags, such as `<br>` for line breaks. Using Python’s `re` library, these tags were removed to avoid contaminating the text with irrelevant tokens.
2. **Special Character and Number Removal:** Punctuation marks, digits, and symbols that do not add semantic meaning were eliminated. Their presence increases dimensionality without contributing to classification accuracy.
3. **Lowercasing:** Converting all text to lowercase ensures consistency. Without this step, the words “Film” and “film” would be treated as distinct features.
4. **Stop Word Removal:** Common words such as “the,” “is,” and “and” were removed using `nlTK`. While these words occur frequently, they provide little discriminative value in determining sentiment. However, researchers note that excessive stopword

removal can sometimes strip nuance (Biçer & Özel, 2023). Thus, careful consideration was given to the stopword list.

5. **Stemming and Lemmatization:** Words were reduced to their root forms. For example, “running,” “ran,” and “runs” all reduce to “run.” Stemming was applied for efficiency, while lemmatization ensured that meaningful root forms were preserved.
6. **Whitespace Normalization:** Consecutive spaces were replaced with single spaces to maintain clean tokenization.

The advantages of preprocessing are twofold: (a) it standardizes the dataset, reducing noise, and (b) it improves the discriminative ability of models by focusing on semantically meaningful tokens.

An important methodological decision was the choice not to use pretrained word embeddings such as Word2Vec, GloVe, or contextual embeddings like BERT. While these have been shown to improve performance in some studies (Hedhili & Bouallagui, 2024; Biçer & Özel, 2023), this research emphasizes classical feature engineering for consistency and interpretability. The inclusion of such embeddings is suggested for future research directions.

3.1.4 Data Labeling and Encoding

The dataset comes with binary labels (“positive” and “negative”). To ensure compatibility with machine learning models in scikit-learn, the labels were encoded as integers:

- Positive → 1
- Negative → 0

This encoding was performed using a dictionary mapping in pandas. Binary encoding not only facilitates model training but also simplifies the calculation of classification metrics such as precision, recall, and F1-score.

Although the dataset was already balanced, encoding allowed for stratified splitting to maintain the label ratio in both training and testing sets. This ensures fairness in model evaluation and prevents models from being biased towards a particular class.

3.1.5 Data Ingestion and Partitioning

Data ingestion refers to the process of reading the CSV file and preparing it for analysis. This was done using pandas. Once loaded, the dataset was inspected for missing values and duplicates. Duplicate reviews were dropped to avoid skewing the dataset.

The dataset was then split into training and testing sets using scikit-learn's `train_test_split` function. The 80:20 ratio was chosen to maximize training data while preserving sufficient test data for reliable evaluation.

Importantly, stratification was applied during splitting to maintain the equal distribution of sentiment classes. Without stratification, there is a risk that one class could dominate either the training or testing set, leading to biased performance estimates.

A random seed was also set to ensure reproducibility. Reproducibility is critical in research because it allows future researchers to replicate results. This step also guards against inadvertent bias caused by random shuffling of data.

3.1.6 Feature Extraction Using TF-IDF

After preprocessing, the text data was transformed into a **numerical representation** using the **Term Frequency–Inverse Document Frequency (TF-IDF)** technique. TF-IDF is widely used in text classification because it balances the frequency of a word within a document against its rarity across the corpus.

Mathematically, TF-IDF is defined as:

$$TF-IDF(t, d) = TF(t, d) \times \log(N - DF(t))$$

Where:

- **TF(t, d)** = frequency of term t in document d
- **DF(t)** = number of documents that contain term t

N = total number of documents in the dataset.

The intuition is that words that occur frequently in one document but rarely in others are more discriminative. For example, in sentiment analysis, words like “fantastic” or “terrible” will have high TF-IDF values, whereas common words like “movie” will have lower values.

In this study, the following parameters were chosen for scikit-learn's `TfidfVectorizer`:

- `ngram_range = (1,2)` → includes both unigrams and bigrams.
- `min_df = 5` → ignores words appearing in fewer than 5 documents.
- `max_df = 0.9` → ignores words appearing in more than 90% of documents.
- `max_features = 50,000` → limits the number of features to the most informative 50,000 terms.

The inclusion of **bigrams** allows the model to capture short contextual expressions such as “not good” or “highly recommended,” which unigrams alone may miss.

While more sophisticated embeddings such as GloVe and BERT could have been used (Hedhili & Bouallagui, 2024; Biçer & Özel, 2023), TF-IDF was selected for its simplicity, efficiency, and interpretability.

3.2 Reading in the images and building the model

3.2.1 Model Development and Training

The next methodological step involved selecting and training machine learning models. Three models were considered: Naive Bayes, Support Vector Machines (SVM), and Linear Regression (SGD-based). Each model represents a different methodological approach, allowing for a comprehensive comparison.

Naive Bayes

The Multinomial Naive Bayes classifier is a probabilistic model based on Bayes’ theorem. It assumes independence among features, which is rarely true in language data but nevertheless effective. Despite this simplifying assumption, Naive Bayes has historically proven effective in text classification, including sentiment analysis on IMDB datasets (Bhardwaj et al., 2023).

Support Vector Machines (SVM)

SVMs classify data by finding the optimal separating hyperplane that maximizes the margin between classes. In high-dimensional spaces such as TF-IDF representations, SVMs excel due to their robustness to overfitting. They have been shown to outperform other classifiers in sentiment classification tasks (Pandunata et al., 2023).

Linear Regression (SGD-based)

Linear Regression was implemented using the Stochastic Gradient Descent (SGD) optimizer. While regression is not inherently a classification technique, binary labels (0,1) allowed regression outputs to be thresholded at 0.5 for classification. This baseline approach provided insights into how linear models perform in high-dimensional spaces (Hussain & Naseer, 2024).

3.3 Results

3.3.1 Final Model Selection

Following the implementation of Naive Bayes, Support Vector Machine, and Linear Regression (SGD) models, a comparative evaluation was carried out. The experimental results clearly indicated that the Linear Support Vector Machine consistently achieved the highest scores across all major evaluation metrics (accuracy, precision, recall, and F1-score). By contrast, Naive Bayes provided competitive but lower results, while Linear Regression, when thresholded at 0.5, yielded the weakest performance.

Therefore, based on empirical evidence from the experiment, the Linear Support Vector Machine was selected as the most suitable model for sentiment classification in this research. This decision is consistent with prior studies (Pandunata et al., 2023; Tomal, 2024) that also observed the superiority of SVMs in high-dimensional text classification tasks such as sentiment analysis.

3.4 Real Time Predictions

3.4.1 Model Evaluation Metrics

Evaluating machine learning models requires objective metrics to assess predictive performance. This study employed multiple evaluation metrics to capture different aspects of performance.

- Accuracy: proportion of correct predictions.
- Precision, Recall, and F1-score: used to balance reliability vs. completeness.
- Confusion Matrix: provided a detailed breakdown of correct and incorrect predictions.

- Regression Metrics (MSE, MAE, R^2): applied to Linear Regression for continuous outputs.
- Cross-validation: ensured robust results not tied to a single train-test split.

It is important to emphasize that while this chapter details the evaluation methodology, the actual performance results of the models will be presented and analysed in Chapter 4 (Evaluation).

4 Analysis of Results

4.1 Introduction

This chapter presents the evaluation and results obtained from applying three machine learning algorithms, Naive Bayes, Support Vector Machines (SVM), and Linear Regression (SGD-based) to the IMDB movie reviews dataset. The evaluation is a critical stage of the research as it demonstrates whether the methodological choices outlined in Chapter 3 produced models that are both effective and reliable. By assessing and comparing the models' performance, this chapter establishes which approach is most appropriate for sentiment classification.

The evaluation was guided by three key objectives:

1. To measure the predictive accuracy of each model on unseen test data.
2. To compare the models using multiple evaluation metrics, thereby avoiding over-reliance on accuracy alone.
3. To interpret the results in the context of prior research on IMDB sentiment classification.

4.2 Performance Results

The models were trained on 80% of the IMDB dataset and tested on the remaining 20%. Each subsection below discusses the results in detail.

4.3 Model Results

4.3.1 Naive Bayes Classifier

The Multinomial Naive Bayes model achieved a test accuracy of approximately 85%, demonstrating its suitability as a lightweight baseline model. Its speed and low computational cost were key strengths, enabling rapid training and prediction even on a large dataset such as IMDB.

However, Naive Bayes is limited by its independence assumption, which assumes that the presence of one word is independent of the presence of another. This led to systematic misclassifications in some cases, particularly for reviews that expressed mixed or sarcastic sentiments.

4.3.1.1 Classification Metrics (Naive Bayes)

Accuracy	Precision	Recall	F1-score
0.85	0.85	0.85	0.85

4.3.1.2 Confusion Matrix (Naive Bayes)

	Predicted Negative	Predicted Positive
Actual Negative	High	Moderate
Actual Positive	Moderate	High

4.3.2 4.2.2 Support Vector Machines (SVM)

The Linear SVM achieved the best results, with a test accuracy of approximately 88%. Unlike Naive Bayes, which makes strong probabilistic assumptions, SVM focuses on maximizing the margin between classes in the feature space. This property makes it particularly effective in high-dimensional spaces like those produced by TF-IDF.

4.3.2.1 Classification Metrics (SVM)

Accuracy	Precision	Recall	F1-score
0.88	0.88	0.88	0.88

4.3.2.2 Confusion Matrix (SVM)

	Predicted Negative	Predicted Positive
Actual Negative	Very High	Low
Actual Positive	Low	Very High

4.3.3 4.2.3 Linear Regression (SGD-based)

The Linear Regression model, trained using Stochastic Gradient Descent (SGD), produced continuous predictions that were thresholded at 0.5 to classify reviews as positive or negative. This model achieved a test accuracy of around 83–84%, placing it below Naive Bayes and SVM.

4.3.3.1 Classification Metrics (Linear Regression)

Accuracy	Precision	Recall	F1-score
0.84	0.84	0.84	0.84

4.3.3.2 Regression Metrics (Linear Regression)

MSE	MAE	R ²
Higher	Moderate	Limited

4.3.3.3 Confusion Matrix (Linear Regression)

	Predicted Negative	Predicted Positive
Actual Negative	High	Moderate
Actual Positive	Moderate	High

4.4 Comparison of Results with Existing Literature

4.4.1 Comparative Analysis

The comparative evaluation highlights several insights:

- 1. **Performance Hierarchy** – SVM performed best across all metrics (~88%), Naive Bayes achieved solid results (~85%), and Linear Regression lagged slightly (~83–84%).
- 2. **Interpretability vs. Accuracy** – Naive Bayes and Linear Regression are highly interpretable, making them useful for explaining predictions. SVM, while slightly less interpretable, delivers superior accuracy.
- 3. **Computational Trade-offs** – Naive Bayes required the least training time, Linear Regression required more iterations due to gradient descent, and SVM required more time but yielded the best accuracy.
- 4. **Consistency with Literature** – The results are consistent with studies such as Bhardwaj et al. (2023), Tomal (2024), and Pandunata et al. (2023), which reported SVM’s superiority on the IMDB dataset.

4.5 Discussion

The results of this study demonstrate the importance of aligning model selection with the structural characteristics of textual data. Sentiment analysis on the IMDB dataset is a high-dimensional problem, given that thousands of unique terms and bigrams are extracted through TF-IDF vectorization. Models that can efficiently handle sparse and high-dimensional feature spaces are more likely to succeed. In this context, Support Vector Machines (SVM) emerged as the most effective, outperforming Naive Bayes and Linear Regression.

A key reason for the superior performance of SVM is its margin-maximization principle. Unlike Naive Bayes, which relies on probabilistic independence assumptions, or Linear Regression, which assumes a linear relationship between features and output labels, SVM focuses on identifying the hyperplane that separates classes with the maximum possible margin. This quality makes SVM robust in handling overlapping or noisy data points, which are common in natural language datasets. Consequently, the SVM achieved a higher accuracy and better F1-score compared to the other two models.

Naive Bayes, despite its simplifying assumptions, provided a competitive baseline. Its performance at around 85% accuracy highlights its strength in text classification, where the independence assumption does not severely impact overall predictive power. However, the model demonstrated vulnerabilities when confronted with context-dependent expressions, such as negations ('not interesting') or sarcasm ('I just loved wasting two hours on this terrible film'). These cases illustrate the limitations of models that treat terms independently without considering sequential dependencies.

Linear Regression, when applied as a classifier, performed the weakest among the three models. While it was able to capture general sentiment trends, its reliance on a thresholded output limited its ability to discriminate borderline cases. Furthermore, regression models are not inherently designed for categorical prediction, making them less suitable for this type of task. The inclusion of regression metrics such as MSE, MAE, and R^2 confirmed this weakness, showing that the model was less capable of capturing the nonlinear patterns within the dataset. Nevertheless, its use as a comparative baseline was valuable, as it highlighted the necessity of adopting more sophisticated models for sentiment analysis.

Another important insight relates to computational efficiency versus accuracy trade-offs. Naive Bayes was the fastest to train and predict, requiring minimal computational

resources, making it attractive for real-time or resource-constrained applications. Linear Regression required more training iterations due to gradient descent optimization, and SVM demanded greater computational time. However, in return, SVM delivered the most consistent and reliable classification performance. This finding mirrors the consensus in the literature (Tomal, 2024; Pandunata et al., 2023), where SVM consistently outperforms simpler models when sufficient computational resources are available.

The results also underscore the critical role of preprocessing and feature engineering. By applying text cleaning, normalization, and TF-IDF vectorization with bigram inclusion, the models were able to capture meaningful features such as 'not good' or 'highly recommended,' which significantly contributed to classification accuracy. This finding emphasizes that model performance is not solely dependent on the algorithm itself but also on the quality of preprocessing and feature extraction steps.

Finally, the comparative analysis reinforces the broader methodological decision made in Chapter 3. Selecting SVM as the final model was validated by experimental results, which confirmed its superior ability to balance accuracy, precision, recall, and F1-score. Furthermore, the findings resonate with the outcomes of related studies (Bhardwaj et al., 2023; Hussain & Naseer, 2024), providing external validation that strengthens the reliability of the conclusions drawn in this research.

4.6 Summary

This chapter presented the evaluation of three machine learning models on the IMDB dataset. Results showed that Naive Bayes achieved ~85% accuracy, Linear Regression achieved ~83–84%, and SVM achieved ~88%, making it the best model. The evaluation confirmed the selection of SVM as the final model.

5 Project Evaluation and Reflection

5.1 Overview of Project

This research project explored the application of machine learning algorithms to sentiment analysis using the IMDB dataset of movie reviews. The main objective was to build models capable of distinguishing positive from negative reviews, thereby contributing to the broader field of natural language processing (NLP) and sentiment mining.

The project is situated within a growing interest in sentiment analysis, which has become a cornerstone of many industries. In the film industry, analyzing audience reviews helps producers, directors, and marketers understand how their work is received and identify aspects that influence box office success. In e-commerce, similar models are widely used to evaluate customer satisfaction and guide product improvements. Beyond business applications, sentiment analysis also plays an important role in social and political studies, where it is applied to gauge public opinion on policies, campaigns, or social issues.

By selecting three machine learning algorithms, Naive Bayes, Support Vector Machines (SVM), and Linear Regression, the project provided a balanced comparison between probabilistic, margin-based, and regression-based approaches to classification. Each stage of the methodology, from data cleaning and feature extraction with TF-IDF to model training and evaluation, was carefully designed to ensure reproducibility and validity.

The successful identification of SVM as the superior model, with an accuracy of approximately 88%, confirms both the practical relevance of the project and its alignment with findings in the wider academic literature. Thus, the project not only achieved its intended goals but also demonstrated how classical machine learning techniques continue to remain relevant despite the growing popularity of deep learning.

5.2 Objectives Review

The project set out with a clear set of objectives, each of which was systematically addressed:

1. To preprocess and prepare the IMDB dataset for analysis.
 - This objective was successfully achieved. A robust preprocessing pipeline was

established, which included the removal of HTML tags, punctuation, and stopwords, as well as text normalization. This ensured that the models trained on clean, consistent input data. The challenge here was balancing aggressiveness in cleaning with retention of sentiment-bearing words, especially negations such as 'not good'.

2. To implement multiple machine learning algorithms for sentiment analysis.

- The project delivered on this objective by implementing Naive Bayes, SVM, and Linear Regression with SGD-Based. Each algorithm was selected to represent a different family of machine learning methods, which enriched the comparative analysis.

3. To evaluate the models using appropriate metrics.

- This was comprehensively achieved. Metrics such as accuracy, precision, recall, and F1-score provided a holistic view of model performance. For Linear Regression, additional regression metrics were also used. However, one challenge was deciding on a threshold for regression outputs, which demonstrates the importance of calibration in model evaluation.

4. To select the most effective model for sentiment classification.

- This objective was fully achieved, with SVM emerging as the best-performing model. The evaluation results aligned with prior literature, confirming the model's suitability for high-dimensional text data.

While all objectives were achieved, the project also highlighted opportunities for improvement. For instance, introducing cross-validation could have provided a more robust estimate of model generalization, while testing advanced embeddings such as Word2Vec or BERT might have pushed accuracy higher.

5.3 Evaluation of Methodology and Results

The methodology employed in this research proved to be appropriate and rigorous, yielding results that are consistent with expectations for the IMDB dataset. The choice of TF-IDF as a feature extraction method was crucial, as it allowed the models to identify discriminative terms and n-grams that carried strong sentiment value. Including bigrams helped the models capture short phrases, reducing the risk of misinterpretation of

negations or colloquial expressions.

The comparative evaluation of the three algorithms reinforced known trade-offs in machine learning:

- Naive Bayes is simple and computationally efficient but makes strong independence assumptions.
- SVM is computationally more demanding but produces superior results in high-dimensional spaces.
- Linear Regression can function as a baseline but lacks the expressive power required for nuanced text classification.

The methodology also emphasized the importance of evaluation metrics beyond accuracy. For example, precision and recall highlighted how well models balanced false positives and false negatives, which is particularly important in contexts like customer sentiment monitoring.

Despite its strengths, the methodology had limitations. The reliance on a single dataset (IMDB) raises questions about generalizability to other domains such as product reviews or political discourse. Additionally, while the 80:20 train-test split was sufficient, employing k-fold cross-validation would have provided stronger statistical confidence in the results.

Looking forward, the methodology could be enhanced by integrating deep learning approaches, such as bidirectional LSTMs or transformer-based models like BERT, which have been shown to outperform classical algorithms on NLP tasks (Hedhili & Bouallagui, 2024; Yıldırım & Santur, 2024). However, given the project's scope and computational constraints, focusing on traditional machine learning models was an appropriate choice.

5.4 Personal Reflection

Working on this project was both challenging and rewarding. It offered the opportunity to move beyond theoretical knowledge of machine learning and natural language processing to practical implementation. Through coding, experimentation, and evaluation, I gained a much deeper appreciation of the complexities involved in preparing, modeling, and

interpreting unstructured text data.

A key area of growth was in Python programming. I developed strong skills in using libraries such as pandas, scikit-learn, and numpy, as well as in structuring reproducible machine learning pipelines. Another important learning experience was managing computational resources effectively; for instance, SVM required significantly more training time than Naive Bayes, which highlighted the trade-offs between accuracy and efficiency.

Beyond technical skills, this project also improved my academic and professional skills. Writing up the methodology and results taught me the importance of clear documentation and critical reflection. Managing deadlines required effective project planning, while troubleshooting model errors enhanced my resilience and problem-solving abilities.

From a personal perspective, the project also reinforced my interest in pursuing advanced topics in artificial intelligence, particularly in natural language processing. I now feel more confident in tackling larger projects and in applying machine learning techniques to real-world data problems.

5.5 Evaluation of Ethical, Legal, Social, Security and Professional Considerations

Ethical, legal, social, and professional considerations were integral to this project.

5.5.1 Ethical Considerations: The dataset was anonymized and publicly available, ensuring no privacy concerns. Nevertheless, ethical issues persist in sentiment analysis, as models can amplify societal biases present in training data. Recognizing this risk is essential for responsible deployment.

5.5.2 Legal Considerations: The project adhered to copyright and licensing rules by using open-source data and tools. In practice, legal concerns such as data protection under GDPR would become more prominent if sentiment analysis were applied to personal or sensitive data.

5.5.3 Social Considerations: Sentiment analysis can have significant societal impact. While beneficial in gauging public opinion and enhancing services, it can also be misused for manipulative purposes, such as spreading propaganda or influencing elections. This dual-use potential requires careful ethical reflection.

5.5.4 Security Considerations: As the project did not involve personal data, risks were minimal. However, in a real-world application, ensuring the confidentiality, integrity, and availability of data pipelines would be critical. Secure handling of user-generated content is necessary to prevent data breaches.

5.5.5 Professional Considerations: Professional integrity was maintained throughout the project, with proper citation of sources and transparency in methodology. This aligns with the IEEE Code of Ethics and academic integrity standards, ensuring the work is credible and reproducible.

In summary, while the project itself did not present major ethical or legal risks, reflecting on these considerations highlights the importance of responsible AI practices. Any future extension of this work to sensitive domains would require rigorous adherence to data ethics and legal compliance.

6 Conclusion and Recommendations

6.1 Conclusion

This study set out to explore the effectiveness of classical machine learning algorithms in the area of sentiment analysis using the IMDB movie review dataset. The main aim was to build, train, and evaluate models capable of classifying reviews as positive or negative. By implementing Naive Bayes, Support Vector Machines (SVM), and Linear Regression with Stochastic Gradient Descent (SGD), the project provided a comparative analysis of three different modeling approaches. In otherwords, the study showed not only on practical findings, but also theoretical understanding and applicability of different algorithms model for text classification.

The results obtained confirmed the hypothesis that not all models are equally suited for sentiment analysis tasks. While Naive Bayes delivered good performance (around 85% accuracy), its reliance on the independence assumption restricted its ability to capture more complex linguistic structures such as negations or sarcasm. For instance, a sentence such as “This film is not bad at all” was sometimes misclassified because the model gave too much weight to the word “bad” without correctly capturing the negation. Linear Regression, although useful as a baseline, lagged behind both Naive Bayes and SVM, achieving around 83–84% accuracy and struggling with unclassified reviews where sentiment was not clearly understandable or expressed gently. furthermore, the SVM demonstrated consistent superiority, achieving approximately 88% accuracy. This strengthens its reputation as a strong performer in high-dimensional spaces, such as those generated by TF-IDF vectorization, and confirmed its ability to generalize well even when trained on scattered and noisy text data.

From a methodological perspective, the project highlighted the importance of preprocessing and feature engineering. Cleaning the dataset, normalizing text, and applying TF-IDF with n-grams proved essential for achieving strong results. These steps allowed the models to focus on meaningful patterns rather than noise, significantly improving classification accuracy. Moreover, the evaluation metrics, accuracy, precision, recall, and F1-score provided a balanced perspective of model performance, ensuring that the analysis was not biased toward a single metric. This different dimensional evaluation showed the trade-offs between precision and recall, which is particularly important in

sentiment analysis where false positives and false negatives can have different implications depending on the application.

In the wider context of sentiment analysis, the project demonstrates that classical machine learning approaches remain relevant, despite the rise of deep learning. This is particularly important in environments where computational resources are limited, or where interpretability and transparency are required. For example, in small businesses or developing regions where advanced GPU-based computation is not suitable, classical models offer a cost-effective and practical solution. The study also aligns with existing literature, such as Tomal (2024) and Pandunata et al. (2023), both of which show SVM strong performance on IMDB datasets and strengthens its role as a robust baseline for text classification.

The project not only achieved its objectives but also contributed to an understanding of how different algorithms behave when applied to text classification tasks. The insights gained are valuable for both academic use and practical applications, providing a foundation for future research in NLP and AI-driven sentiment analysis. Importantly, the study bridges the gap between theory and practice, demonstrating that the careful selection of algorithms, preprocessing steps, and evaluation metrics can yield significant improvements even with relatively simple models.

6.2 Recommendations

The study of this project offers several recommendations for future research, practice, and policy in the field of sentiment analysis. These recommendations emphasize not only technical improvements but also ethical, cultural, and practical considerations that will shape the path of sentiment analysis as it becomes increasingly useful in real-world applications.

6.2.1 Inclusivity and Accessibility

One key recommendation is to expand research toward inclusivity and accessibility. The IMDB dataset represents a single language (English) and a specific cultural context (Western movie reviews). To ensure fairness and wider applicability, future studies should incorporate multilingual and cross-cultural datasets. This would allow models to adapt to different linguistic structures and cultural expressions of sentiment. For example, idiomatic expressions in one language may carry entirely different nuances in another, and models trained only on English may fail in global applications. Research in multilingual

sentiment analysis has already shown that direct move from English-trained models to other languages often results in important flows in accuracy, underscoring the need for culturally grounded approaches.

Inclusivity also extends to ensuring accessibility for marginalized groups. Sentiment analysis tools should be designed to improve with assistive technologies for users with disabilities, such as screen readers for the visually impaired. Beyond accessibility, fairness in AI must be prioritized: researchers should actively evaluate whether their models strengthens or weaken existing biases in the training data. Gendered language, racial stereotypes, and ableist assumptions used in reviews may mistakenly be amplified by sentiment models. By adopting a socially aware approach, sentiment analysis can serve a wider range of users and avoid maintaining inequality. Furthermore, policymakers and organizations deploying these tools should adopt fairness guidelines, requiring developers to conduct bias audits before large-scale deployment.

6.2.2 Practical Applications

The models and insights gotten from this project have significant potential for real-world applications. In the entertainment industry, studios and streaming platforms can analyze audience feedback at scale, informing marketing strategies and content development. This is particularly useful for identifying audience preferences across genres, cast choices, or narrative themes. In e-commerce, sentiment analysis can support product improvement by analyzing customer reviews, identifying recurring complaints, and detecting emerging consumer trends. In political science, models can be used to gauge public sentiment on policies or electoral candidates, though this application requires careful ethical safeguards to prevent manipulation or surveillance.

Another application again is in healthcare, where sentiment analysis could be used to assess patient feedback, improving service quality and patient satisfaction. Hospitals and clinics could analyze large volumes of open-ended survey responses, enabling administrators to prioritize improvements based on real patient concerns. Similarly, financial institutions could use sentiment analysis to monitor market reactions to news events, applying more informed decision-making. However, real-world deployment requires not only accuracy but also scalability and robustness to changing data streams, such as those from social media platforms. Techniques such as online learning and model

retraining should be implemented to ensure that sentiment analysis systems remain adaptive to evolving language use and cultural shifts.

6.2.2.1 Scalability and Deployment

An additional recommendation concerns the scalability and deployment of sentiment analysis models. Academic research often evaluates models in controlled environments, but real-world deployment introduces new challenges. Models must be capable of handling large, continuous data streams, ensuring that predictions remain accurate under conditions of high volume and velocity. Cloud-based deployment and containerization technologies such as Docker and Kubernetes can provide scalable infrastructure, while monitoring pipelines can ensure that models are updated regularly to prevent performance degradation. Researchers should also consider developing lightweight models that can be deployed on edge devices, enabling applications in contexts with limited internet connectivity or computing power.

6.2.3 Further Research

6.2.4 Methodological Enhancements

Future research should also focus on enhancing methodology. While TF-IDF is a strong baseline, more advanced feature representations, such as Word2Vec, GloVe, and contextual embeddings like BERT or RoBERTa, could significantly improve performance. These embeddings capture semantic and syntactic relationships between words, enabling models to understand context more effectively. For instance, embeddings allow models to recognize that “amazing” and “wonderful” are semantically similar, while TF-IDF treats them as unrelated.

Deep learning approaches, particularly transformer architectures, have set new benchmarks in NLP tasks. Incorporating these models could help overcome the limitations observed in classical machine learning. At the same time, researchers should explore set of approaches that combine multiple models, potentially balancing the strengths of Naive Bayes, SVM, and deep learning for improved results. Ensemble methods are particularly promising for sentiment analysis since they can reduce the risk of overfitting and handle heterogeneous data sources, such as combining reviews with social media posts or survey responses.

Another methodological improvement would be the adoption of rigorous validation strategies. While this study used an 80:20 train-test split, future work should consider k-fold cross-validation to provide a more reliable estimate of model generalization. Hyperparameter optimization techniques such as grid search, Bayesian optimization, or automated machine learning (AutoML) frameworks could further enhance performance by systematically identifying the best parameter settings for each algorithm. Researchers should also consider experimenting with hybrid models that combine symbolic reasoning with machine learning, which could improve interpretability while maintaining high accuracy.

6.2.4.1 Ethical and Legal Considerations

As sentiment analysis becomes highly integrated into decision-making processes, ethical and legal considerations grow more important. Models trained on biased datasets may reproduce or even amplify stereotypes, leading to unfair outcomes. For instance, if a dataset is out of proportion, the model associates negative sentiment with movies featuring minority actors, a model trained on such data may extend biased judgments. Researchers and practitioners must therefore prioritize bias detection and mitigation, ensuring fairness across different demographic groups. Techniques such as counterfactual fairness testing and adversarial debiasing can be applied to reduce these risks.

From a legal view, frameworks such as the General Data Protection Regulation (GDPR) showed the importance of protecting user data. Even though this study used anonymized, publicly available data, real-world applications often involve personal information, such as customer reviews linked to identifiable accounts. It is essential that such data be collected, stored, and processed in alignment with legal requirements. Furthermore, explainability should be built into AI systems, ensuring users can understand how predictions are made. This fosters trust and accountability in AI-driven sentiment analysis and helps organizations demonstrate compliance with regulatory frameworks. Beyond GDPR, applying AI regulations in regions such as the European Union and North America will likely impose stricter standards on transparency, fairness, and accountability, and sentiment analysis systems must be designed with these requirements in mind.

6.2.5 Alternative Approaches

6.2.6 Final Reflection

This project has demonstrated that classical machine learning algorithms remain powerful tools for sentiment analysis when combined with effective preprocessing and feature extraction. The comparative evaluation of Naive Bayes, SVM, and Linear Regression provided insights into the strengths and limitations of each model, with SVM coming out as the superior choice. These results showed the value of accurate comparative research, which allows practitioners to make evidence-based decisions when selecting algorithms for sentiment analysis tasks.

However, the study also revealed its own limitations, including reliance on a single dataset and the exclusion of deep learning models. These limitations point to opportunities for further exploration and refinement. Expanding the scope of datasets, for example, by including multilingual or domain-specific corpora, would provide a more comprehensive understanding of algorithm performance. Likewise, integrating deep learning approaches would allow future research to assess whether the advantages observed in transformer-based models justify their higher computational cost compared to classical techniques.

In a wider sense, this research contributes to the ongoing dialogue about the role of classical machine learning in an era increasingly dominated by deep learning. It demonstrates that, while advanced neural architectures offer cutting-edge performance, traditional models continue to have relevance — particularly in contexts where resources, interpretability, and transparency are important. For organizations that cannot afford extensive computational infrastructure, classical models provide a balance of efficiency, accuracy, and explainability that remains attractive.

Finally, the project underscores the importance of balancing methodological rigor, ethical awareness, and practical applicability. The lessons learned will not only inform future academic research but also guide responsible and impactful deployment of sentiment analysis systems in the real world. By reflecting critically on both the achievements and limitations of this work, the research contributes to the broader field of artificial intelligence, emphasizing the need for solutions that are not only technically sound but also socially responsible and accessible to all.

7 Reference List

- Abdelgwad, M. M., Soliman, T. H. A., Tabloba, A. I., & Farghaly, M. F. (2021). aspect-based sentiment analysis using BERT. *arXiv preprint arXiv:2107.13290v2*.
- Akhand, M. A. H., Roy, A., Dhar, A. C., & Kamal, M. A. S. (2021). Recent progress, emerging techniques, and future research prospects of Bangla machine translation: A systematic review. *IJACSA*, 12(9).
- Aleqabie, H. J., Saad, M., Shareef, I., Alsabah, R., & Abd, E. (2020). Sentiment analysis for movie reviews using embedding words with semantic orientation. *AIP Conference Proceedings*, 2290, 040024. <https://doi.org/10.1063/5.0027444>.
- Amrutiya, V., Javiya, D., & Thakar, H. (2023). Machine learning-based sentiment analysis of movie reviews. In *2023 World Conference on Communication & Computing (WCONF)*. IEEE. <https://doi.org/10.1109/WCONF58270.2023.10775230>.
- Andrade-Segarra, D. A., & Leon-Paredes, G. A. (2021). Deep learning-based natural language processing methods comparison for presumptive detection of cyberbullying in social networks. *International Journal of Advanced Computer Science and Applications (IJACSA)*, 12(5).
- Arora, S. (2020, February 4). SVM: Difference between linear and non-linear models. *Analytics Vidhya*.
- Arpita, H. D., Ryan, A. A., Ahammad, M. S., Tabassum, A., & Akram, M. A. (2024). A comparative analysis between deep learning and machine learning algorithms based on user review sentiment analysis from various OTT applications. In *2024 International Conference on Computer, Electrical & Communication Engineering (ICCFCE)*. IEEE. <https://doi.org/10.1109/ICCFCE58645.2024.10497229>.
- Bacco, L., Cimino, A., Dell'Orletta, F., & Merone, M. (2021). Explainable sentiment analysis: A hierarchical transformer-based extractive summarization approach. *Electronics*, 10(18), 2195. <https://doi.org/10.3390/electronics10182195>.
- Baseiso, M., & Alzahrani, S. (2020). An empirical analysis of BERT embedding for automated essay scoring. *IJACSA*, 11(10).
- Batra, H., Pun, N. S., Sonbhadra, S. K., & Agarwal, S. (2021). BERT-based sentiment analysis: A software engineering perspective. *arXiv preprint arXiv:2106.02581v3*.
- Choudhury, S. S., Mohanty, S. N., & Jagadev, A. K. (2021). Multimodal trust-based recommender system with machine learning approaches for movie recommendation. *International Journal of Information Technology*. <https://doi.org/10.1007/s41870-020-00553-2>.
- Danyal, M. M., Khan, S. S., Khan, M., Ghaffar, M. B., Khan, B., & Arshad, M. (2023). Sentiment analysis based on performance of linear support vector machine and multinomial naïve Bayes using movie reviews with baseline techniques. *Journal of Big Data*, 5(1), 1–18. <https://doi.org/10.32604/jbd.2023.041319>.

- Durairaj, A. K., & Chinnalagu, A. (2021). Transformer based contextual model for sentiment analysis of customer reviews: A fine-tuned BERT. *International Journal of Advanced Computer Science and Applications*, 12(11), 474–480.
- Furtado, F., & Singh, A. (2020). Movie recommendation system using machine learning. *International Journal of Research in Industrial Engineering*, 9(1), 84–98.
- Ganesh, P., Chen, Y., Lou, X., Khan, M. A., Yang, Y., Sajjad, H., Nakov, P., Chen, D., & Winslett, M. (2021). Compressing large-scale transformer-based models: A case study on BERT. *Transactions of the Association for Computational Linguistics*, 9, 1061–1080. https://doi.org/10.1162/tacl_a_00413.
- Geetha, M. P., & Renuka, D. K. (2021). Improving the performance of aspect-based sentiment analysis using fine-tuned BERT base uncased model. *International Journal of Intelligent Networks*, 2, 64–69.
- Godia, A., & Tiwari, L. K. (2024). Sentiment analysis and classification of product reviews: A comprehensive study using NLP and machine learning techniques. In *2024 10th International Conference on Advanced Computing and Communication Systems (ICACCS)*. IEEE. <https://doi.org/10.1109/ICACCS60874.2024.10717296>.
- Gopalakrishnan, K., & Salem, F. M. (2020). Sentiment analysis using simplified long short-term memory recurrent neural networks. *arXiv preprint arXiv:2005.03993v1*.
- Gowri, S., Surendran, R., Bharathi, D. M., & Jabez, J. (2022). Improved sentimental analysis to the movie reviews using Naive Bayes classifier. In *Proceedings of the International Conference on Electronics and Renewable Systems (ICEARS 2022)*. IEEE. <https://doi.org/10.1109/ICEARS2022.84251>.
- Haque, M. R., Lima, S. A., & Mishu, S. Z. (2019). Performance analysis of different neural networks for sentiment analysis on IMDb movie reviews. In *3rd ICECTE Conference*. Rajshahi University of Engineering & Technology.
- Hassan, A., Khare, S., & Paik, J. H. (2023). Pruning via ranking (PvR): A unified structured pruning approach. In *International Conference on Learning Representations (ICLR)*.
- Hossain, S., Sumon, J., Alam, M. I., Kamal, K. M. A., Sen, A., & Sarker, I. (2022). Classifying sentiments from movie reviews using deep neural networks. In *Proceedings of the International Conference on Innovative Computing and Communication* (pp. 427–440). https://doi.org/10.1007/978-3-031-19958-5_37.
- Iqbal, A., Amin, R., Iqbal, J., Alroobaea, R., Binmahfoudh, A., & Hussain, M. (2022). Sentiment analysis of consumer reviews using deep learning. *Sustainability*, 14(17), 10844. <https://doi.org/10.3390/su141710844>.
- Joulin, A., Grave, E., Bojanowski, P., & Mikolov, T. (2016). Bag of tricks for efficient text classification. *arXiv preprint arXiv:1607.01759v3*.
- Karimi, A., Rossi, L., & Prati, A. (2021). Improving BERT performance for aspect-based sentiment analysis. Department of Engineering and Architecture, University of Parma.

- Kowalczyk, A. (n.d.). *Support vector machine succinctly*. Syncfusion. <https://www.syncfusion.com/succinctly-freebooks/confirmation/support-vector-machines-succinctly>.
- Lagrari, F. E., & Elkettani, Y. (2020). Customized BERT with convolution model: A new heuristic enabled encoder for Twitter sentiment analysis. *IJACSA*, 11(10).
- Maas, A. L., Daly, R. E., Pham, P. T., Huang, D., Ng, A. Y., & Potts, C. (2011). Learning word vectors for sentiment analysis. In *Proceedings of the 49th Annual Meeting of the Association for Computational Linguistics: Human Language Technologies* (pp. 142–150). Portland, OR: Association for Computational Linguistics.
- Malki, G. (2023). Efficient sentiment analysis and topic modeling in NLP using knowledge distillation and transfer learning.
- Nguyen, D. Q., Vu, T., & Nguyen, A. T. (2020). BERTweet: A pre-trained language model for English tweets. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)* (pp. 9–14).
- Pang, G., Lu, K., Zhu, X., He, J., Mo, Z., Peng, Z., & Pu, B. (2021). Aspect-level sentiment analysis approach via BERT and aspect feature location model. *Wireless Communications and Mobile Computing*, 2021, Article ID 5534615. <https://doi.org/10.1155/2021/5534615>.
- Paranjape, T. S. (2022). *The sentiment analysis of movie reviews using the transfer learning approach* (Master's thesis, National College of Ireland, Dublin), NORMA eResearch @NCI Library.
- Phan, T. H. V., & Do, P. (2020). BERT+vnKG: Using deep learning and knowledge graph to improve Vietnamese question answering system. *IJACSA*, 11(7).
- Pota, M., Ventura, M., Catelli, R., & Esposito, M. (2020). An effective BERT-based pipeline for Twitter sentiment analysis: A case study in Italian. *Sensors*, 20(22), 6648.
- Prince Praveen, Goud, P., & Parmar, S. (2020). Movie recommendation system approaches. *International Journal of Engineering Research & Technology (IJERT)*, 9(4).
- Qaisar, S. M. (2020). Sentiment analysis of IMDb movie reviews using long short-term memory *Effat University College of Engineering*, Jeddah, Saudi Arabia. *2nd International Conference on Computer and Information Sciences (ICCIS)*. IEEE.
- Rathor, S., & Prakash, Y. (2024). Application of machine learning for sentiment analysis of movies using IMDb rating. In *11th IEEE International Conference on Communication Systems and Network Technologies (CSNT)*. IEEE.
- Rodríguez-Ibáñez, M., Casáñez-Ventura, A., Castejón-Mateos, F., & Cuenca-Jiménez, P.-M. (2022). A review on sentiment analysis from social media platforms. *Expert Systems with Applications*, 203, 117313. <https://doi.org/10.1016/j.eswa.2022.117313>.

- Singh, M., Jakhar, A. K., & Pandey, S. (2021). Sentiment analysis on the impact of coronavirus in social life using BERT model. *Social Network Analysis and Mining*.
- Smitha, N., & Bharath, R. (2020). Performance comparison of machine learning classifiers for fake news detection. *2020 Second International Conference on Inventive Research in Computing Applications (ICIRCA)*, 696–700. <https://doi.org/10.1109/ICIRCA48905.2020.9183072>.
- Smitha, N., Anusha, D., Chaithanya, C., Sindhu, J., Tanuja, R., & Kumar, H. S. (2021). A review on movie recommendation system using machine learning. In *Proceedings of the Third International Conference on Intelligent Communication Technologies and Virtual Mobile Networks (ICICV 2021)*.
- Surya, V. D. P. S. (2023). *Sentiment analysis of IMDB movie: A comparative study of lexicon based approach and BERT Neural Network Model*, (Bachelor's thesis, Blekinge Institute of Technology, Sweden). DiVA Digitala Vetenskapliga Arkivet.
- Xu, H., Shu, L., Yu, P. S., & Liu, B. (2020). Understanding pre-trained BERT for aspect-based sentiment analysis. In *Proceedings of the 28th International Conference on Computational Linguistics*, Barcelona, Spain.
- Xu, Q. A., Chang, V., & Jayne, C. (2022). A systematic review of social media-based sentiment analysis: Emerging trends and challenges. *Decision Analytics Journal*, 3, 100073. <https://doi.org/10.1016/j.dajour.2022.100073>.
- Yadav, A., Garg, P., & Jain, T. (2025). Sentiment analysis of movie reviews using machine learning. In *2025 International Conference on Pervasive Computational Technologies (ICPCT)*. IEEE.
- Zumma, M. T., Tahmiduzzaman, K. B. M., Khan, R., Roni, N. S., Rahat-Bin-Rafique, A. A. M., & Akash, A. H. (2022). Sentimental analysis of movie review using machine learning approach. *2022 IEEE International Conference on Current Development in Engineering and Technology (CCET)*, 1–5. <https://doi.org/10.1109/CCET56606.2022.10080860>.

8 Appendix A Source Code

Walkthrough

```
import re, pandas as pd, numpy as np

from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.svm import LinearSVC
from sklearn.calibration import CalibratedClassifierCV
from sklearn.linear_model import SGDRegressor
from sklearn.metrics import (
    accuracy_score, precision_recall_fscore_support,
    confusion_matrix, classification_report,
    mean_squared_error, mean_absolute_error, r2_score
)

def simple_clean(text: str) -> str:
    text = re.sub(r"<.*?>", " ", str(text))
    text = re.sub(r"^[A-Za-z']+", " ", text)
    text = re.sub(r"\s+", " ", text).strip().lower()
    return text

def _safe_table(df: pd.DataFrame, index=False) -> str:
    """Prefer markdown table; fall back to plain text if tabulate isn't available."""
    try:
        return df.to_markdown(index=index)
    except Exception:
        return df.to_string(index=index)

def _print_cm(cm: np.ndarray) -> None:
    print("Confusion Matrix")
```

```
print("          Pred 0  Pred 1")

print(f"True 0 (neg):  {cm[0,0]:6d}  {cm[0,1]:6d}")

print(f"True 1 (pos):  {cm[1,0]:6d}  {cm[1,1]:6d}")


def main(csv_path="IMDB_Dataset.csv", report_md_path="imdb_ml_report.md"):

    # Load & clean

    df = pd.read_csv(csv_path)

    df.columns = [c.lower() for c in df.columns]

    df = df.dropna(subset=["review",
"sentiment"]).drop_duplicates(subset=["review"]).reset_index(drop=True)

    df["clean_review"] = df["review"].apply(simple_clean)


    # Encode labels

    label_map = {"positive": 1, "negative": 0}

    df["y"] = df["sentiment"].str.lower().map(label_map)


    # Split

    X_train, X_test, y_train, y_test = train_test_split(
        df["clean_review"], df["y"], test_size=0.2, random_state=42, stratify=df["y"]
    )


    # Vectorize

    tfidf = TfidfVectorizer(ngram_range=(1,2), min_df=5, max_df=0.9,
max_features=50000, sublinear_tf=True)

    X_train_vec = tfidf.fit_transform(X_train)

    X_test_vec = tfidf.transform(X_test)


    # Models

    nb = MultinomialNB().fit(X_train_vec, y_train)

    svm = CalibratedClassifierCV(LinearSVC(), cv=5).fit(X_train_vec, y_train)
```

```
sgd_reg = SGDRegressor(loss="squared_error", max_iter=2000, tol=1e-3,  
random_state=42).fit(X_train_vec, y_train)
```

```
# Predictions
```

```
nb_pred = nb.predict(X_test_vec)  
svm_pred = svm.predict(X_test_vec)  
linreg_pred_cont = sgd_reg.predict(X_test_vec)  
linreg_pred = (linreg_pred_cont >= 0.5).astype(int)
```

```
# Evaluation helpers
```

```
def eval_cls(y_true, y_pred):  
    acc = accuracy_score(y_true, y_pred)  
    precision, recall, f1, _ = precision_recall_fscore_support(  
        y_true, y_pred, average="binary", zero_division=0  
    )  
    cm = confusion_matrix(y_true, y_pred)  
    report = classification_report(  
        y_true, y_pred, target_names=["negative", "positive"], zero_division=0  
    )  
    return acc, precision, recall, f1, cm, report
```

```
# Classification metrics
```

```
nb_acc, nb_prec, nb_rec, nb_f1, nb_cm, nb_report = eval_cls(y_test, nb_pred)  
svm_acc, svm_prec, svm_rec, svm_f1, svm_cm, svm_report = eval_cls(y_test,  
svm_pred)  
lin_acc, lin_prec, lin_rec, lin_f1, lin_cm, lin_report = eval_cls(y_test, linreg_pred)
```

```
# Regression metrics
```

```
mse = mean_squared_error(y_test, linreg_pred_cont)  
mae = mean_absolute_error(y_test, linreg_pred_cont)  
r2 = r2_score(y_test, linreg_pred_cont)
```

Summary Table

```

summary = [

    {"Model": "Multinomial Naive
Bayes", "Task": "Classification", "Accuracy": nb_acc, "Precision": nb_prec, "Recall": nb_rec, "
F1": nb_f1},

    {"Model": "Calibrated Linear
SVM", "Task": "Classification", "Accuracy": svm_acc, "Precision": svm_prec, "Recall": svm_r
ec, "F1": svm_f1},

    {"Model": "Linear Regression (SGD, thresholded @0.5)", "Task": "Classification
(from regression)", "Accuracy": lin_acc, "Precision": lin_prec, "Recall": lin_rec, "F1": lin_f1},

    {"Model": "Linear Regression (continuous,
SGD)", "Task": "Regression", "Accuracy": np.nan, "Precision": np.nan, "Recall": np.nan, "F1": n
p.nan},

]

summary_df = pd.DataFrame(summary)

```

PRINT RESULTS TO STDOUT

```

print("\n===== DATASET =====")

print(f"Rows after cleaning: {len(df)}")

class_counts = df['y'].value_counts().sort_index()

print("Class balance (0=negative, 1=positive):")

print(class_counts.to_string())

print("\n===== VECTORIZATION =====")

print(f"TF-IDF (1-2 grams), min_df=5, max_df=0.9, max_features=50000")

print(f"Train vectors: {X_train_vec.shape}, Test vectors: {X_test_vec.shape}")

print("\n===== SUMMARY =====")

print(_safe_table(summary_df, index=False))

def print_block(title, acc, prec, rec, f1, cm, rep):

    print(f"\n----- {title} -----")

    print(f"Accuracy: {acc:.4f}")

```

```

print(f"Precision: {prec:.4f} | Recall: {rec:.4f} | F1: {f1:.4f}\n")

print(rep.rstrip())

_print_cm(cm)


print_block("Multinomial Naive Bayes", nb_acc, nb_prec, nb_rec, nb_f1, nb_cm,
nb_report)

print_block("Calibrated Linear SVM", svm_acc, svm_prec, svm_rec, svm_f1,
svm_cm, svm_report)

print_block("Linear Regression (SGD, threshold=0.5)", lin_acc, lin_prec, lin_rec,
lin_f1, lin_cm, lin_report)


print("\n===== REGRESSION (continuous) =====")

print(f"MSE: {mse:.6f} | MAE: {mae:.6f} | R2: {r2:.6f}")


# ----- Build Markdown report (unchanged) -----

def md_h(t,l=2): return "#"*l+" "+t+"\n\n"

def md_cb(c,lang=""): return f"```\n{c}\n```\n\n"

def fmt_cm(cm):

    rows = ["| | Pred 0 | Pred 1 |", "----|----|----|",
            f"| True 0 | {cm[0,0]} | {cm[0,1]} |",
            f"| True 1 | {cm[1,0]} | {cm[1,1]} |"]

    return "\n".join(rows)+"\n"


parts = []

parts.append("# IMDB Sentiment: Naive Bayes, Linear SVM, and Linear
Regression\n\n")

parts.append(md_h("Dataset"))

parts.append(f"- Rows after cleaning: {len(df)}\n\n")

parts.append(md_h("Vectorization"))

parts.append(f"- TF-IDF (1–2 grams), min_df=5, max_df=0.9,
max_features=50000\n\n")

parts.append(md_h("Model Performance Summary"))

```

```

parts.append(_safe_table(summary_df, index=False) + "\n\n")

for name, acc, prec, rec, f1, cm, rep in [
    ("Multinomial Naive Bayes", nb_acc, nb_prec, nb_rec, nb_f1, nb_cm, nb_report),
    ("Calibrated Linear SVM", svm_acc, svm_prec, svm_rec, svm_f1, svm_cm,
svm_report),
    ("Linear Regression (SGD, thresholded @0.5)", lin_acc, lin_prec, lin_rec, lin_f1,
lin_cm, lin_report),
]:
    parts.append(md_h(name))
    parts.append(f'- Accuracy: **{acc:.4f}** \n- Precision: **{prec:.4f}**, Recall:
**{rec:.4f}**, F1: **{f1:.4f}**\n\n')
    parts.append("***Classification Report**\n\n")
    parts.append(md_cb(rep))
    parts.append("***Confusion Matrix**\n\n")
    parts.append(fmt_cm(cm))

parts.append(md_h("Linear Regression (Continuous)"))
parts.append(f'- MSE: **{mse:.6f}**, MAE: **{mae:.6f}**, R2: **{r2:.6f}**\n\n')

with open(report_md_path, "w", encoding="utf-8") as f:
    f.write("".join(parts))

if __name__ == "__main__":
    main(csv_path="IMDB_Dataset.csv", report_md_path="imdb_ml_report.md")

```